

Peer-to-Peer Directory Synchronization with Lightweight Version Control

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June 22, 2026

Declaration of Own Work

Required. Print this page and sign it in front of the advisor on presentation day.

This work has been composed solely by us. This work has not been accepted in any previous application for a degree or diploma. The work of which this is a record has been wholly done by us. All extracts, including those created by generative AI, have been distinguished by quotation marks and the sources of our information have been specifically acknowledged.

Date: _____

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Acknowledgements

TODO: Each member writes a line or two. Thank the advisor, EPITA, each other, etc.

Abstract

Write this last, at the same time as the Conclusions. One paragraph about the problem, one about the solution. More about the problem than the solution.

TODO: Write abstract.

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Chapter 1

Introduction

Template says: “It is often best to write this section at the same time you write the conclusion.”

1.1 Motivations

Why should the reader care? Why is this research important?

Students need to share project files. Current tools either need servers, require technical knowledge, or lack version safety. We want something simpler.

TODO: Arihant: expand with concrete examples from our coursework experience.

1.2 Context of the Project

Context: who we are, what programme, what the project is about.

Action Learning project at EPITA. Four-person team. Two backend (Go, C++), two frontend/CI-CD (Java, GitHub Actions).

TODO: All: confirm team role descriptions.

1.3 Objectives

What the research is trying to achieve.

Three objectives:

1. Design a serverless P2P sync system with automatic peer discovery.
2. Provide version safety through automatic backups before overwrites.
3. Validate with two hypotheses: H_1 (latency), H_2 (consistency).

1.4 Overview of the Thesis

One sentence per chapter saying what it covers.

TODO: Write after all chapters are done.

Chapter 2

Literature Review

Template says: “The literature review should be used to develop a working framework and shared vocabulary.” It is NOT a descriptive list of tools. It is NOT a summary of what you read. You must analyze and add value.

Mention how research was divided among team members. Move from general to specific.

2.1 The Collaborative File Management Landscape

Brief overview of four categories: centralized, VCS, P2P sync, local sharing. One paragraph each, focused on what we learned, not feature lists.

TODO: All: review and confirm the categories and assessments are fair.

2.2 Critical Analysis of Synchronization Approaches

This is the core section. Discuss what the literature taught us and how it changed our design. Include disagreements, limitations, open questions.

2.2.1 The Ordering Problem

Lamport clocks vs. vector clocks. Why we chose Lamport. What we lose.

2.2.2 Consistency vs. Availability

CAP theorem. Why we chose availability. What “eventually consistent” means for us.

2.2.3 Conflict Resolution

LWW vs. CRDTs vs. operational transformation. Why we chose LWW with backups.

2.2.4 Efficient Data Propagation

rsync algorithm, epidemic algorithms. Why we start with whole-file transfers.

TODO: Arihant/Huizhe: these are mostly done. Review for accuracy.

2.3 Identified Gap

Show that no existing tool covers all five dimensions well. Reference the comparison tables in the Appendix.

TODO: All: verify the gap summary table is accurate.

2.4 Positioning the Project

Key differentiators: LAN-only, zero config, built-in backups, split architecture.

2.5 Summary

Three main takeaways from the literature and how they shaped the design.

Chapter 3

State of the Art and Design Choices

Template says: “Describe the state of the art resulting from the research you conducted. Include the decisions and choices you made for technology, architecture, design, methodology. Document all relevant sources with citations. Briefly describe and critically assess cited works.”

3.1 Summary

TODO: Write a summary paragraph after the sections below are done.

3.2 Zero-Configuration Peer Discovery

mDNS/DNS-SD. Why we chose it. Limitations (multicast filtering).

3.3 Event Ordering and Logical Clocks

Lamport clocks. Known limitation (cannot detect concurrency). Trade-off vs. vector clocks.

3.4 Consistency Model

CAP theorem application. Eventually consistent. Backup as safety net.

3.5 Conflict Resolution

LWW with local backups. Based on Bayou. Schema ready for CRDTs later.

3.6 Other Technical Decisions

TODO: Add sections for: bandwidth (rsync future), file watching (inotify/FSEvents), hashing (SHA-256), transport (TCP).

Chapter 4

System Architecture

Template says: “This chapter describes the design adopted by this research to achieve the aims and objectives stated in the Introduction.”

4.1 System Architecture Anticipated

Submitted with thesis.

4.1.1 Components

Three components: Go daemon, C++ daemon, Java frontend.

TODO: All: add a diagram here. Even a simple box diagram helps.

4.1.2 Data Flow

Step-by-step sync cycle.

4.1.3 Metadata Storage

SQLite schema.

4.1.4 Communication Protocols

mDNS, TCP, IPC.

TODO: Arihant/Huizhe: fill in protocol details. Nizar/Shengkang: fill in frontend-to-backend interface.

4.2 System Architecture Updated

Left blank. Completed at end of Action Learning Week.

Chapter 5

Methodology

Template says: “Discuss the methodology, the stages of implementation, and the research design. Detail participants, instruments, procedure, data analysis.”

This is where the action plan fits.

5.1 Methodology Anticipated

Submitted with thesis.

5.1.1 Team Structure

Who does what.

5.1.2 Development Process

Four phases: research, prototyping, integration, evaluation.

5.1.3 Tools

Languages, build systems, CI/CD, test environment.

5.1.4 Research Design

Two hypotheses. How we test them. What counts as success.

5.1.5 Action Plan

Timeline table: Pre-ALW, Day 1–2, Day 3–4, Day 5.

TODO: All: review the action plan timeline. Is it realistic?

5.2 Methodology Updated

Left blank. Completed at end of Action Learning Week.

Chapter 6

Results

6.1 Development Anticipated

Submitted with thesis. What we plan to build.

Expected outputs: Go daemon, C++ daemon, Java frontend, hypothesis test results.

TODO: All: list specific deliverables for your component.

6.2 Development Updated

Left blank with thesis. Submitted separately at end of ALW. Maximum half a page.

Chapter 7

Conclusions

Template says: “Summarize the problem, the solution, and its main innovative features, outlining future work.”

TODO: Write at the same time as the Abstract.

APA style required. Use in-text citations throughout. **TODO:** All: make sure every claim has a citation. Every source in the bibliography must be cited in the text.

Appendix A

Comparison Tables

Detailed dimension-by-dimension comparison tables moved here from Chapter 2 to keep the literature review concise.

TODO: All: verify ratings in the tables are accurate and fair.

Appendix B

Flesch Readability Statistics

Include Flesch Reading Ease and Flesch-Kincaid Grade Level statistics here as required by the coordinator.

TODO: All: compute and include readability stats after final text edits.

Appendix C

List of Figures

TODO: Generated automatically once we add figures.

Appendix D

List of Tables

TODO: Generated automatically once we add tables.